SOFTWARE REQUIREMENTS SPECIFICATION

Waste Tracking Operations System for Resource Work Cooperative

Abstract

This Software Requirements Specification (SRS) defines the requirements for the Waste Tracking Operations System being developed for Resource Work Cooperative. The system aims to digitize selected waste tracking processes that are currently managed using paper forms and Excel spreadsheets. The proposed solution will provide simple web-based data entry, structured database storage, record management, and basic reporting functions. The system will focus on the main operational workflows that can reasonably be developed, tested, and deployed within the internship period.

Thaveesha Kuruppuarachchi

IT Intern- Indus Institute

Version	Date	Changes
0.1	05/10/2026	Initial SRS draft.
0.2	13/10/2026	Updated requirements after staff clarification.
0.3	14/10/2026	Added system diagrams, technology stack, and updated requirements.

Table of Contents

List of Tables iv

List of Figures iv

1. Introduction 1

1.1. Purpose 1

1.2. Project Background 1

1.3. Project Scope 1

1.4. Out of Scope 2

1.5. Definitions and Terminology 3

2. Current System Overview 4

2.1. Current Manual Process 4

2.2. Current Operational Workflows 4

2.2.1 Crate and Pallet Recording 4

2.2.2 Salvage Recording 5

2.2.3 Electrical Crate and Pallet Recording 5

2.2.4 Pickup Recording 5

2.2.5 Whitegoods Recording 6

2.2.6 Collectable Recording 6

2.2.7 Electrical Testing Recording 7

2.2.8 E-Waste Pickup Recording 7

2.2.9 Scrap and Outbound Material Recording 8

2.3. Current Data Recording Tools 8

2.4. Problems with Current System 9

2.5. Current Process Diagram 10

2.6. Current System Limitations 10

3. Proposed System Overview 11

3.1. System Objective 11

3.2. Main System Functions 11

3.3. User Types 12

3.4. Proposed Workflow 13

3.5. System Context 13

3.6. Proposed System Boundaries 14

3.7. System Technologies 15
4. Functional Requirements 15
4.1. User Access 15
4.2. Crate and Pallet Recording 16
4.3. Salvage Recording 16
4.4. Electrical Crate and Pallet Recording 17
4.5. Pickup Recording 17
4.6. Whitegoods Recording 18
4.7. Collectable Recording 18
4.8. Electrical Testing Recording 18
4.9. E-Waste Pickup Recording 19
4.10. Scrap Recording 19
4.11. Record Management 20
4.12. Reporting and Dashboard 20
4.13. General Form Behaviour 21
5. Data Requirements 21
5.1. Data to be Stored 21
5.2. Material and Container Data 22
5.3. Validation and Calculations 23
5.4. Data Storage and History 23
6. Business Rules 24
7. Non-Functional Requirements 25
7.1. Usability 26
7.2. Device Support 27
7.3. Security 27
7.4. Reliability and Data Protection 27
7.5. Maintainability 28
7.6. System Downtime and Operational Continuity 28
7.7. Testing and User Training 28
8. System Constraints and Assumptions 29
8.1. Project Constraints 29
8.2. Current Assumptions 29

8.3. Scope Control 30
9. Acceptance Criteria 30
9.1. Minimum Successful Delivery 31
10. Future Enhancements 32

List of Tables

Table 1: Definitions and Terminology 3
Table 2: Proposed System Technologies 15
Table 3:Functional Requirements - User Access 15
Table 4: Functional Requirements - Crate and Pallet Recording 16
Table 5: Functional Requirements - Salvage Recording 16
Table 6: Functional Requirements - Elec. Crate Pallet Recording 17
Table 7: Functional Requirements -Pickup Recording 17
Table 8: Functional Requirements - Whitegoods Recording 18
Table 9: Functional Requirements - Collectables Recording 18
Table 10: Functional Requirements -Electrical Testing 18
Table 11: Functional Requirements - E-waste Pickup Recording 19
Table 12: Functional Requirements - Scrap Recording 19
Table 13: Functional Requirements - Record Management 20
Table 14: Functional Requirements - Reporting & Dashboard 20
Table 15: Functional Requirements -General Forms 21
Table 16: Data Storing Logic 22
Table 17: Container Tare Weights 23
Table 18: Business Rules 24
Table 19: Non-Functional Requirements 25
Table 20: Project Constraints 29
Table 21: Assumptions 29
Table 22: Acceptance Criteria 30

List of Figures

Figure 2.5-1: Current AS-IS Process Diagram 10
Figure 3.5-1: Proposed System Context Diagram 13
Figure 3.5-2: High-Level Use Case Diagram 14

1. Introduction

1.1. Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for the Waste Tracking Operations System being developed for Resource Work Cooperative.

The system will support the digitization of selected waste tracking and operational processes that are currently recorded using paper forms and Excel spreadsheets. The document will provide a clear reference for the development, testing, deployment, and future improvement of the system.

This SRS will be updated during the project as further requirements are confirmed with staff and management.

1.2. Project Background

Resource Work Cooperative receives donated and recovered items from different sources, including customer drop-offs, household and council pickups, and salvage operations at council waste facilities.

Items that can be reused are processed for resale, while materials that cannot be resold may be recycled, scrapped, or sent to landfill. Waste and material quantities are currently recorded using several paper datasheets. The recorded information is later manually entered into Excel spreadsheets for storage, calculation, and reporting.

This process requires repeated manual data entry and depends heavily on paper records and spreadsheet-based processing.

The proposed Waste Tracking Operations System will digitize the main data collection processes by allowing staff to enter operational data directly into a web-based system. The data will be stored in a structured database and used for reporting and waste tracking purposes.

1.3. Project Scope

The project will focus on developing a simple, mobile-friendly web application for recording and managing the main waste tracking activities carried out by Resource Work Cooperative.

The initial scope includes:

- Crate and pallet recording
- Salvage recording
- Electrical crate and pallet recording
- Pickup recording
- Whitegoods recording
- Collectable recording
- Electrical testing data recording
- E-waste pickup recording
- Scrap and outbound material recording
- Viewing and managing stored records
- Basic waste tracking reports and dashboards

The system will replace or reduce the need for manual paper forms and repeated manual entry into Excel spreadsheets.

The project will focus on the main operational workflows and will not attempt to digitize every process within Resource Work Cooperative.

1.4. Out of Scope

The following areas are not included in the initial version of the system unless they are later identified as essential:

- Detailed tracking of individual donated items
- Full resale inventory management
- Detailed electrical repair and maintenance tracking
- Direct integration with weighing equipment
- Upload and storage of weighbridge dockets or supporting documents
- Advanced staff KPI and performance tracking
- Automated regulatory or government submissions
- Advanced reporting, forecasting, and analytics beyond the required operational summaries
- Complex role-based access control beyond basic Staff and Manager/Admin access

These functions may be considered as future improvements.

1.5. Definitions and Terminology

Table 1: Definitions and Terminology

Term	Description
Crate/Pallet	A container used to collect and move incoming materials before they are weighed and processed.
Salvage Up	Waste or unusable material taken from Resource Work Cooperative to a council waste facility for disposal.
Salvage Down	Useful material collected from a council waste facility and brought back to Resource Work Cooperative.
Pickup	Material collected by Resource Work Cooperative from a customer, organization, or council location.
Collectable	An item or group of items identified as having higher resale value and recorded separately when required.
Whitegoods	Large household appliances such as refrigerators, washing machines, dryers, and dishwashers.
E-waste	Electrical and electronic equipment received or collected by Resource Work Cooperative.
Tare Weight	The weight of an empty container, crate, pallet, or similar item that may be deducted from the total measured weight.
Material Category	A classification used to identify the type of material in a load, such as timber, textiles, metals, plastics, or mixed materials.
Waste Diversion	Material prevented from being sent to landfill through reuse, recovery, recycling, or resale.

2. Current System Overview

2.1. Current Manual Process

Resource Work Cooperative currently uses a combination of paper forms and Excel spreadsheets to record waste tracking information.

During daily operations, staff record information such as weights, material categories, percentages, container types, pickup details, salvage activity, electrical items, and whitegoods. Different paper forms are used depending on the type of activity.

After the paper forms are completed, the information is later entered manually into Excel spreadsheets by management. The spreadsheets are then used to store the records, perform calculations, and prepare summary information and reports.

The current process can be summarized as:

Operational activity → Paper form → Manual Excel entry → Data processing → Reporting

Because the same information is first written on paper and later entered again into Excel, the current process involves repeated manual data entry.

2.2. Current Operational Workflows

The main waste tracking workflows identified so far are described below.

2.2.1 Crate and Pallet Recording

General incoming materials are placed into crates or pallets based on their size and handling requirements.

Once a crate or pallet is full, staff:

- weigh the full container
- select the container type
- visually estimate the percentage of each material category
- record the information on the Crate and Pallet Datasheet

The existing form records container type, weight, and material categories such as metals, textiles, timber, building material, paper/cardboard, plastics, glass, and mixed materials.

The material contained in one crate or pallet may come from one customer or several customers. The current process does not require individual donors to be identified.

Electrical items are separated from this general material flow.

2.2.2 Salvage Recording

The salvage process includes two activities within the same trip.

First, Resource Work Cooperative takes unusable material to a council waste facility. This material is recorded as Salvage Up or waste to landfill.

After the waste is unloaded, staff collect reusable material from the council waste facility and bring it back to Resource Work Cooperative. This is recorded as Salvage Down or salvaged material.

The existing Salvage Datasheet records separate information for waste sent to landfill and material recovered from the facility. It includes fields for crew, time, weight, and material categories.

The recorded weights are based on vehicle weight differences measured at the council waste facility.

2.2.3 Electrical Crate and Pallet Recording

Electrical items are separated from general incoming materials and taken to a separate electrical processing area.

The items are placed into designated crates or pallets.

When the container is full, staff:

- record the date
- select the container type
- record the weight
- estimate the percentage breakdown

Electrical material is currently divided into two categories:

- TVs, Monitors and Projectors
- Everything Else

The existing Electrical Crate and Pallet Datasheet uses these two categories.

2.2.4 Pickup Recording

Resource Work Cooperative collects material from external locations, including residential properties, organizations, and council-related locations.

After a pickup is completed, staff record:

- date
- pickup location

- net material weight
- material percentage breakdown

The weight entered represents the final material weight of the collected load.

The existing paper form contains a Delivery ($) field; however, this information is not required in the new system's pickup recording.

The system does not need to separately classify the pickup as residential, organization, or council pickup. The location field is sufficient for the current version.

2.2.5 Whitegoods Recording

Whitegoods include large household appliances such as:

- refrigerators and freezers
- washing machines
- dryers
- dishwashers

Whitegoods are recorded as a monthly quantity rather than as individual weighted items.

For reporting purposes, the total estimated whitegoods weight will be calculated using an average weight of 48 kg per whitegood.

For example:

Estimated Whitegoods Weight = Number of Whitegoods × 48 kg

Whitegoods will continue to follow the Whitegoods process rather than being recorded through other waste tracking workflows.

2.2.6 Collectable Recording

Collectables are items considered to have higher resale value.

Collectables are recorded separately only when they are picked up directly from the outside or driveway area and have not previously been included in another waste tracking record.

If a collectable is found later in a process where its weight has already been recorded, such as the pricing table or electrical testing area, it must not be recorded again as a collectable.

This rule prevents the same material from being counted more than once.

2.2.7 Electrical Testing Recording

Electrical items are tested after they enter the electrical processing area.

Staff use the Electrical Testing Data Sheet to record counts against different electrical item categories and testing or stock statuses.

The current statuses include:

- Untested / Broken / Failed
- Tested and Priced
- Stock Cull
- Sold Last Month
- Diverted to OLS
- Diverted to Coll

Stock Cull means an item that did not sell in the shop and is removed from shop stock and sent to scrap.

Diverted to OLS means the item has been moved to online sales.

Diverted to Coll means the item has been moved to collectables.

Staff currently add a tally each time an electrical item is tested or processed. The future system will replace this manual tally with simple counter-based data entry.

Electrical testing information is for internal operational use only and does not directly contribute to waste reporting.

The system will store counts only and will not track individual electrical items or their full repair history.

2.2.8 E-Waste Pickup Recording

Resource Work Cooperative also performs dedicated e-waste pickups.

The e-waste pickup process currently records information such as:

- date
- organization or source
- total weight
- container or pallet weight

The organization or source of the pickup should be recorded.

After the material is collected, it may later enter the normal Electrical Crate and Pallet process.

The detailed e-waste pickup workflow and the method used to avoid double-counting in reporting are still pending further clarification

2.2.9 Scrap and Outbound Material Recording

Scrap material is generally distributed to relevant collectors approximately once per month.

The current workbook records scrap-related material weights. Current material categories include:

- Iron and Steel
- Aluminium
- Other Non-Ferrous Metals
- Mixed Metals
- E-waste

Scrap income does not need to be recorded in the new system.

The detailed fields required for the scrap process are still pending further clarification and review of the existing scrap form.

Other outbound waste processes, such as cardboard or other specialist waste collection, are outside the current system scope unless later confirmed as required.

2.3. Current Data Recording Tools

The current process uses several different paper forms and spreadsheets.

The main forms identified are:

- Salvage Statistics Datasheet
- Crate and Pallet Datasheet
- Collectable Tub Datasheet
- Pickup Datasheet
- Electrical Crate and Pallet Datasheet
- Electrical Testing Data Sheet

These forms collect information such as:

- dates
- weights
- material percentages
- container types
- locations
- crew details

- times
- quantities
- electrical item counts

After this information is recorded manually, it is entered into the existing waste tracking Excel workbook.

The workbook contains separate data-entry areas and record lists for different processes such as salvage, crates, pallets, electrical material, pickups, whitegoods, scrap, and e-waste pickups.

The workbook also performs additional calculations and reporting.

The exact reporting formulas and final reporting requirements will be reviewed separately later in the project.

2.4. Problems with Current System

The current process requires staff to record information on paper forms and then enter the same information again into Excel spreadsheets. This creates duplicate work and takes extra time.

There is also a risk of data entry errors when information is transferred from paper to Excel.

Because records are stored across different paper forms and spreadsheets, it can be difficult to find and manage historical information.

The current process also depends heavily on spreadsheet formulas and manual data processing for reporting.

Overall, the current system works, but it is time-consuming and relies too much on manual handling of data.

2.5. Current Process Diagram

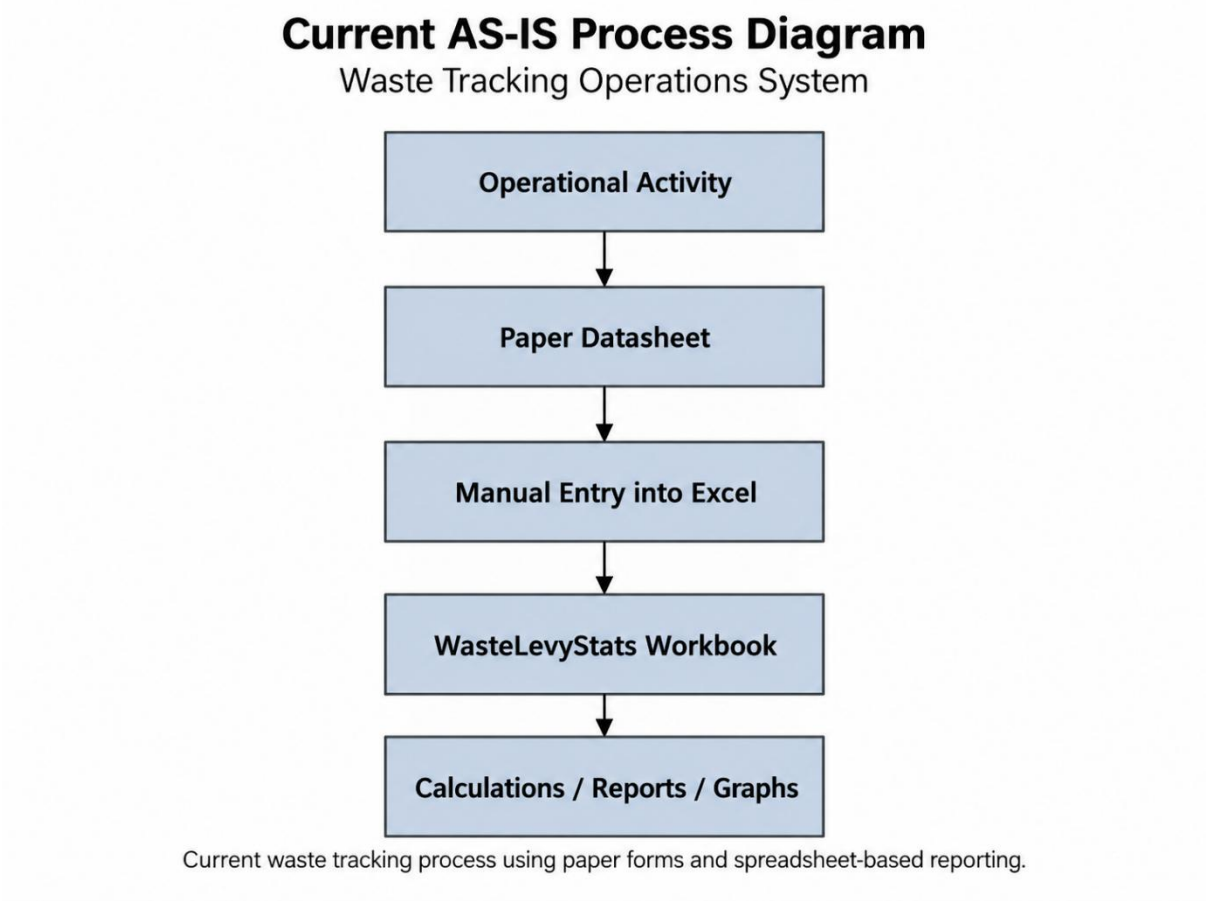


Figure 2.5-1: Current AS-IS Process Diagram

2.6. Current System Limitations

The existing system is functional, but it depends heavily on manual work.

The main limitation is that operational data is not entered directly into a central system when the activity occurs.

The proposed system will aim to reduce this manual process by allowing staff to enter information directly through a web application and store the information in a structured database.

The goal is not to completely replace every existing business process. The initial system will focus only on the main waste tracking activities that can reasonably be completed within the internship project.

3. Proposed System Overview

The proposed Waste Tracking Operations System will provide a simple web-based solution for recording and managing the main waste tracking activities at Resource Work Cooperative.

The system will allow staff to enter operational data directly through a mobile-friendly interface instead of first recording the information on paper and later entering it into Excel.

The entered data will be stored in a central database and made available for basic reporting and management purposes.

The system will focus on the main workflows that can reasonably be completed within the internship period. It will not attempt to replace every process used by Resource Work Cooperative.

3.1. System Objective

The main objective of the system is to reduce manual data entry and make waste tracking information easier to record, store, and access.

The system should allow staff to record the main operational activities using simple digital forms and allow management to review stored records and basic reports.

The system should also provide a structure that can be expanded later if Resource Work Cooperative decides to add more functions.

3.2. Main System Functions

The proposed system will support the following main functions:

- Record crate and pallet data
- Record salvage data
- Record electrical crate and pallet data
- Record pickup data
- Record whitegoods counts
- Record collectable data
- Record electrical testing data
- Record e-waste pickup data
- Record scrap or outbound material data
- View and manage previously entered records
- Provide basic summaries and reports

Some functions may be updated during the project as further requirements are confirmed with staff.

3.3. User Types

The system is expected to have two main user types.

1. **Staff User**

Staff users will mainly use the system to enter operational data.

General staff will use a shared Staff account to access the system. This keeps the login process simple and avoids requiring each staff member to maintain individual login details.

Staff users will be able to:

- create waste tracking records
- review entered information before submission
- correct information before final submission
- view relevant records where permitted

Once a record has been confirmed and submitted to the database, normal staff users will not be able to edit or delete it.

2. **Manager / Administrator**

Manager or Administrator users will have broader access to the system.

They will be able to:

- enter operational data
- view stored records
- edit submitted records
- delete incorrect records
- view reports and summaries
- perform basic administrative functions

Each Manager/Administrator will have an individual account so that administrative actions, such as editing or deleting submitted records, can be linked to the specific user who performed the action.

The system will keep permissions simple and will not require complex role-based access control.

3.4. Proposed Workflow

The future process will aim to remove the unnecessary step of entering the same information twice.

Staff will enter the information directly into the web application at the time of the activity or shortly after it occurs.

The system will then store the information in the database and make it available for later viewing, editing, and reporting.

3.5. System Context

At a high level, the system will connect operational staff and management to one shared source of waste tracking data.

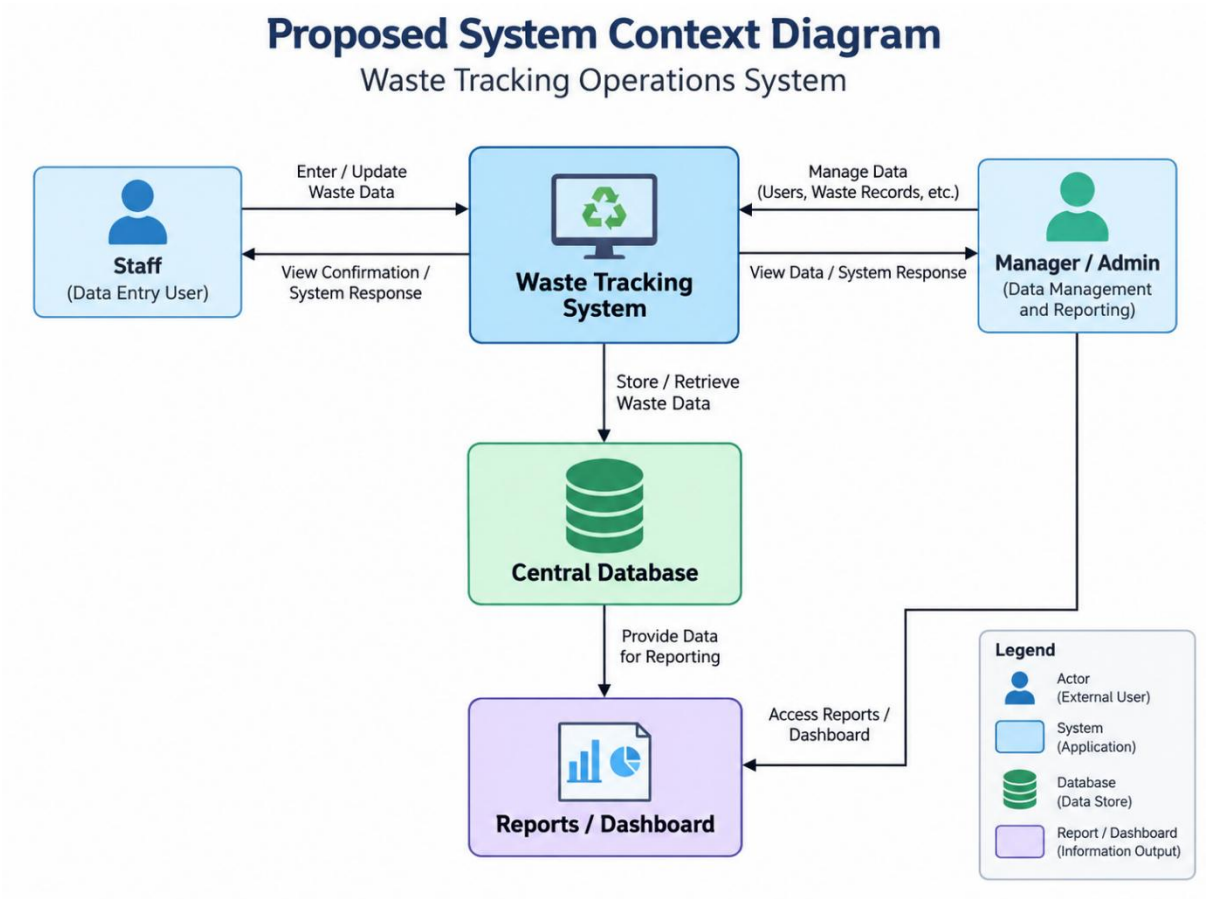


Figure 3.5-1: Proposed System Context Diagram

The system will be designed to work through a web browser and should be suitable for desktop and tablet use.

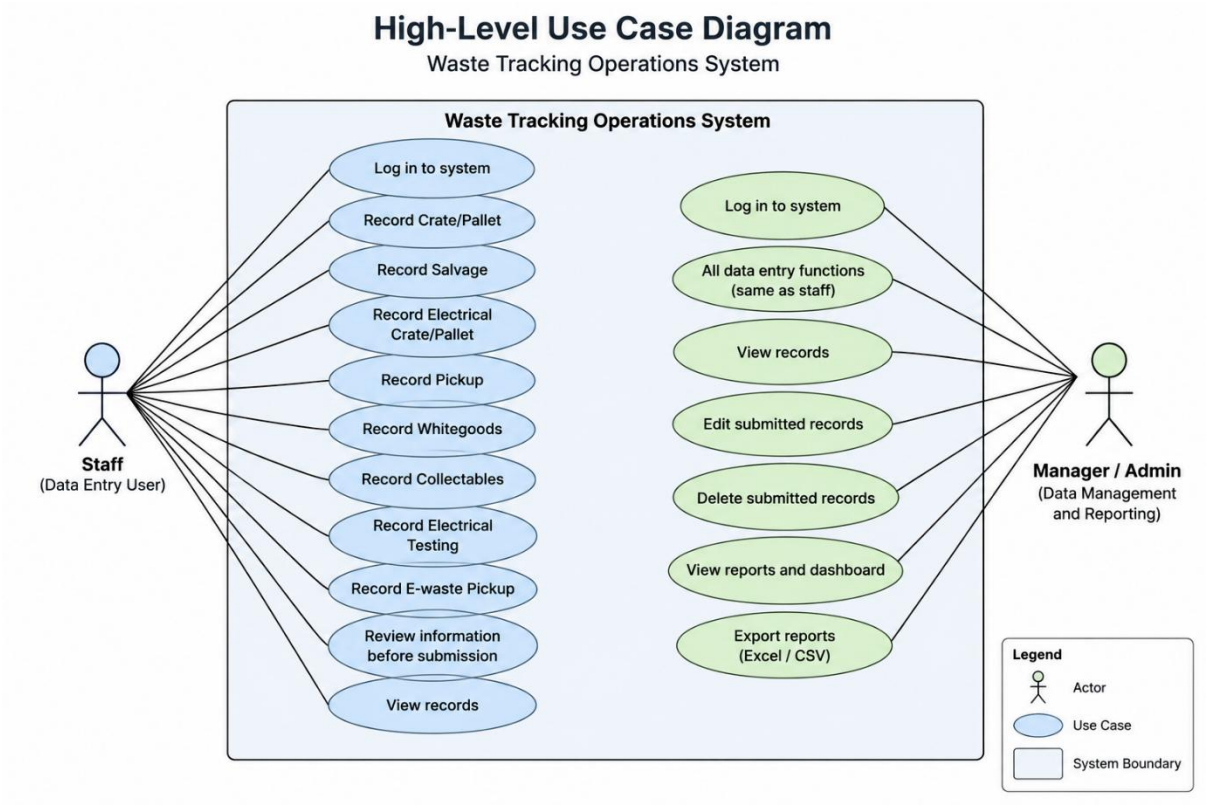


Figure 3.5-2: High-Level Use Case Diagram

3.6. Proposed System Boundaries

The system will focus on the main waste tracking and related operational processes identified for this project.

The initial system will not attempt to provide:

- detailed individual item inventory management
- detailed electrical repair tracking
- direct integration with weighing equipment
- complex staff KPI management
- advanced reporting or forecasting beyond the required operational reporting
- complex role-based access control

Where a process is still unclear or cannot reasonably be completed within the internship period, it may be simplified or moved to future development.

The main priority is to deliver a complete and usable waste tracking system within the project time available.

3.7. System Technologies

The system will use a simple web-based technology stack selected based on development experience, maintainability, cost, and suitability for Resource Work Cooperative.

Table 2: Proposed System Technologies

Area	Technology
Frontend	React with TypeScript
Backend	Java with Spring Boot
Database	PostgreSQL
Hosting / Deployment	Proposed: Google Cloud Platform (GCP), estimated at approximately AUD $40 per month
Reporting	Web-based dashboard with Excel/CSV export
Version Control	Git and GitHub

4. Functional Requirements

This section describes the main functions the Waste Tracking Operations System should provide.

The requirements below represent the current understanding of the system. Some requirements may be refined during development as remaining assumptions are confirmed.

4.1. User Access

Table 3:Functional Requirements - User Access

ID	Functional Requirement
FR-01	The system shall allow authorized users to log in and log out.
FR-02	The system shall support two user roles: Staff and Manager/Administrator.
FR-03	General operational staff shall use a shared Staff account, and each Manager/Administrator user shall have an individual account.
FR-04	Staff users shall be able to enter operational data, while Manager/Administrator users shall also

	be able to manage submitted records and reports.

4.2. Crate and Pallet Recording

Table 4: Functional Requirements - Crate and Pallet Recording

ID	Functional Requirement
FR-05	The system shall allow staff to create a crate or pallet record with date, container type, gross weight, and material percentage breakdown.
FR-06	The system shall automatically deduct the selected container tare weight to calculate net material weight.
FR-07	The system shall check that the material percentages total 100%.
FR-08	The system shall calculate the weight of each material category from the net material weight and entered percentages.
FR-09	The system shall save the completed crate or pallet record in the database.

4.3. Salvage Recording

Table 5: Functional Requirements - Salvage Recording

ID	Functional Requirement
FR-10	The system shall allow staff to record a Salvage activity containing both Salvage Up and Salvage Down information.
FR-11	The system shall require date, crew, Time Up, Kg Up, and material percentage breakdown for Salvage Up.
FR-12	The system shall require Time Down, Kg Down, and material percentage breakdown for Salvage Down.
FR-13	The system shall check that each material percentage breakdown totals 100%.
FR-14	The system shall accept the final Kg Up and Kg Down material weights entered by staff and shall not calculate vehicle weight differences.

FR-15	The system shall save the completed salvage record in the database.

Council weighbridge dockets do not need to be uploaded or stored in the system

4.4. Electrical Crate and Pallet Recording

Table 6: Functional Requirements - Elec. Crate Pallet Recording

ID	Functional Requirement
FR-16	The system shall allow staff to record electrical crate or pallet data including date, container type, gross weight, and percentage breakdown.
FR-17	The system shall automatically deduct the selected container tare weight.
FR-18	Electrical material shall be divided into TVs/Monitors/Projectors and Everything Else.
FR-19	The system shall check that the two percentages total 100% and calculate the weight of each category.
FR-20	The system shall save the completed electrical crate or pallet record.

4.5. Pickup Recording

Table 7: Functional Requirements -Pickup Recording

ID	Functional Requirement
FR-21	The system shall allow staff to record pickup date, location, net material weight, and material percentage breakdown.
FR-22	The system shall check that pickup material percentages total 100%.
FR-23	The system shall calculate the weight of each material category from the entered pickup weight and percentages.
FR-24	The system shall save the completed pickup record.

4.6. Whitegoods Recording

Table 8: Functional Requirements - Whitegoods Recording

ID	Functional Requirement
FR-25	The system shall allow staff or management to record the number of whitegoods received for the relevant reporting period.
FR-26	The system shall calculate estimated whitegoods weight using 48 kg per recorded whitegood.
FR-27	The system shall save the whitegoods count and calculated weight for reporting.

4.7. Collectable Recording

Table 9: Functional Requirements - Collectables Recording

ID	Functional Requirement
FR-28	The system shall allow staff to record collectable date, measured weight, and material percentage breakdown.
FR-29	The system shall deduct the 2 kg Collectable Tub tare weight where required.
FR-30	The system shall check that material percentages total 100% and calculate material-category weights.
FR-31	The system shall save the collectable record.
FR-32	Collectables shall only be recorded when their weight has not already been captured through another process.

4.8. Electrical Testing Recording

Table 10: Functional Requirements -Electrical Testing

ID	Functional Requirement
FR-33	The system shall provide a simple counter-based interface for electrical testing.
FR-34	Staff shall be able to increment counts for the relevant electrical item category

	and status when an item is tested or processed.
FR-35	The system shall support the existing testing/status categories, including Untested/Broken/Failed, Tested and Priced, Stock Cull, Sold Last Month, Diverted to OLS, and Diverted to Coll.
FR-36	The system shall store electrical testing counts for internal operational use only.

4.9. E-Waste Pickup Recording

Table 11: Functional Requirements - E-waste Pickup Recording

ID	Functional Requirement
FR-37	The system shall allow staff to record e-waste pickup date, organisation/source, total weight, and container/pallet weight.
FR-38	The organisation or source shall be recorded for each e-waste pickup.
FR-39	The system shall save e-waste pickup records in the database.

4.10. Scrap Recording

Table 12: Functional Requirements - Scrap Recording

ID	Functional Requirement
FR-40	The system shall provide a simple function for recording scrap data, including date or reporting period, material type, and weight.
FR-41	The system shall save scrap records in the database.
FR-42	The system shall not require scrap income to be recorded.

4.11. Record Management

Table 13: Functional Requirements - Record Management

ID	Functional Requirement
FR-43	The system shall allow staff to review and correct entered data before final submission.
FR-44	Once confirmed, the record shall be stored in the database.
FR-45	Users shall be able to view previously entered records where permitted.
FR-46	Records should be filterable by date and process type.
FR-47	Only Manager/Administrator users shall be able to edit or delete submitted records.
FR-48	The system shall record the individual Manager/Administrator account responsible for editing or deleting submitted records

4.12. Reporting and Dashboard

Table 14: Functional Requirements - Reporting & Dashboard

ID	Functional Requirement
FR-49	The system shall provide summary reports based on stored waste tracking data.
FR-50	Manager/Administrator users shall be able to view both summary information and detailed operational records.
FR-51	Reports should support filtering by relevant criteria such as date range, material category, process type, source, or location.
FR-52	The system should support relevant reporting periods such as daily, weekly, monthly, financial year, and custom date range.
FR-53	The system should reproduce relevant existing WasteLevyStats graphs where practical.
FR-54	Report data shall be exportable in Excel or CSV format.

FR-55	The system should support generation of the Monthly NRE report.
FR-56	The Manager/Administrator dashboard shall provide a daily operational summary showing the number of records entered for key activities such as Crate/Pallet, Salvage, Pickup, Electrical Crate/Pallet, Collectables, and other relevant workflows.

The dashboard will initially provide simple daily summaries of the main operational activities. For example, management should be able to view how many Crate/Pallet, Salvage, Pickup, and other records were completed during the day. Additional summary information, statistics, and visualizations may be added during development as the reporting requirements and available data become clearer.

4.13. General Form Behaviour

Table 15: Functional Requirements -General Forms

ID	Functional Requirement
FR-57	Required fields shall be clearly identified.
FR-58	The system shall prevent invalid values such as negative weights, quantities, or percentages.
FR-59	The system shall display clear success and error messages when records are submitted.
FR-60	The system should minimise unnecessary typing by using dropdowns, number fields, buttons, and simple selection controls where possible.

5. Data Requirements

The system will store operational waste tracking data in a structured database so that records can be viewed, managed, calculated, and used for reporting.

5.1. Data to be Stored

The system should store the main data required by each workflow.

Table 16: Data Storing Logic

Data Type	Examples
Date and time	Entry date, Salvage Time Up, Salvage Time Down
Weight	Gross container weight, net material weight, pickup weight, salvage weight
Container type	Apple Crate, Stillage Cage, Plastic Pallet, Pine Pallet, Hardwood Pallet
Material category	Timber, metals, textiles, plastics, mixed materials, e-waste
Material percentage	Estimated percentage of each material category
Calculated material weight	Weight calculated from net weight and percentage
Location/source	Pickup location or e-waste pickup organization
Crew information	Salvage crew
Quantity/count	Whitegoods and electrical testing counts
User information	User who created or modified a record
Process type	Crate/Pallet, Salvage, Pickup, Electrical, Collectable, etc.

Not every workflow will use every field.

5.2. Material and Container Data

The system should support the material categories currently used in the existing forms.

Common categories include:

- Iron and Steel
- Aluminium
- Other Non-Ferrous Metals
- Mixed Metals
- Textiles
- Mattresses
- Timber
- Bricks, Blocks and Pavers
- Building and Demolition Material
- Paper, Cardboard and Books
- Plastics
- Glass
- Mixed Materials
- E-waste

Electrical Crate and Pallet records use:

- TVs, Monitors and Projectors
- Everything Else

The system should support the following container tare weights:

Table 17: Container Tare Weights

Container	Tare Weight
Apple Crate	60 kg
Stillage Cage	50 kg
Plastic Pallet	6 kg
Pine Pallet	18 kg
Hardwood Pallet	30 kg
Collectable Tub	2 kg

5.3. Validation and Calculations

The system should apply basic validation before records are saved.

This includes:

- required fields should not be empty
- weights and quantities should not be negative
- percentages should not be negative
- material percentages should total 100% where required
- dates should be valid

Where required, the system shall calculate:

Net Material Weight = Gross Weight – Container Tare Weight

Material category weight shall be calculated as:

Material Weight = Net Material Weight × Material Percentage

Whitegoods estimated weight shall be calculated as:

Whitegoods Weight = Whitegoods Count × 48 kg

5.4. Data Storage and History

Each completed operational entry should be stored as a separate record in the database.

The stored data should allow authorized users to:

- view previous records
- identify the workflow or process type

- identify whether a record was created through the shared Staff account or an Administrator account and record the individual Administrator responsible for administrative changes where applicable.
- edit submitted records where permitted
- use the data for reports and summaries

After final submission, editing and deletion should be restricted to Manager/Administrator users.

Operational records should remain available unless management later defines a specific retention period.

6. Business Rules

This section describes the main rules that should be followed when waste tracking data is entered and processed by the system.

Table 18: Business Rules

ID	Business Rule	Status
BR-01	Material percentage breakdowns must total 100% where percentages are required.	Confirmed
BR-02	Where material is weighed together with a crate, pallet, cage, or tub, the system shall deduct the configured tare weight to calculate net material weight.	Confirmed
BR-03	Collectable, electrical and any other material must not be recorded if its weight has already been captured through another process.	Confirmed
BR-04	Salvage Up represents waste taken from Resource Work Cooperative to a council waste facility for disposal.	Confirmed
BR-05	Salvage Down represents useful material recovered from a council waste facility and brought back to Resource Work Cooperative.	Confirmed
BR-06	Staff will enter the final Kg Up and Kg Down values provided by the council. The system will not calculate truck weight differences.	Confirmed
BR-07	Crew, Time Up, and Time Down are mandatory for Salvage records.	Confirmed
BR-08	Electrical Crate and Pallet material is divided into TVs/Monitors/Projectors and Everything Else.	Confirmed
BR-09	Whitegoods are recorded as counts and converted to estimated weight using 48 kg per whitegood.	Confirmed
BR-10	Pickup weight is treated as the final net material weight of the collected load.	Confirmed

BR-11	The Pickup process does not require Delivery ($) or a separate pickup-type field.	Confirmed
BR-12	Electrical testing information is recorded as counts only and does not require individual item tracking.	Confirmed
BR-13	Electrical testing data is internal operational data and does not directly contribute to waste reporting.	Confirmed
BR-14	Staff material percentage estimates are stored as entered. The system checks the total but does not determine whether the estimate itself is correct.	Confirmed
BR-15	Staff may correct data before final submission. After submission, only Manager/Admin users may edit or delete records.	Confirmed
BR-16	Where the same material passes through multiple internal processes, the first recorded process should be used for final waste/diversion reporting to reduce double-counting.	Confirmed
BR-17	The original source of material should be retained where required for reporting.	Confirmed
BR-18	Reporting calculations will use data stored in the system database. Detailed waste diversion and NRE calculations will be finalised after further analysis of the current Excel workbook.	Pending further analysis

7. Non-Functional Requirements

This section describes how the system should perform and the general quality requirements it should meet.

The project will focus on a simple and reliable solution that is practical for Resource Work Cooperative and can be completed within the internship period.

Table 19: Non-Functional Requirements

ID	Requirement Area	Requirement
NFR-01	Usability	The system should be simple and easy to use for staff with different levels of technical experience.
NFR-02	Usability	Data-entry forms should use clear labels, dropdown lists, number fields, and simple controls where possible.
NFR-03	Mobile Compatibility	The system should work on common desktop, tablet web browsers.
NFR-04	Responsive Design	The interface should adjust to different screen sizes so staff can use the system on site.

NFR-05	Performance	Normal form submissions, record viewing, and basic searches should complete within a reasonable time under normal use.
NFR-06	Reliability	Data entered successfully should be stored correctly and remain available for later use.
NFR-07	Reliability	The system should avoid creating duplicate records when a form is submitted.
NFR-08	Security	Only authorized users should be able to access the system.
NFR-09	Security	User passwords, if used, should be stored securely and should not be stored as plain text.
NFR-10	Access Control	Manager or Administrator functions should only be available to authorized users.
NFR-11	Data Integrity	The system should apply basic validation before saving records.
NFR-12	Maintainability	The system should use a simple structure and readable code so it can be maintained or extended later.
NFR-13	Maintainability	Basic setup, deployment, and system usage should be documented.
NFR-14	Backup	Operational data should be backed up regularly using the selected hosting or database platform.
NFR-15	Availability	The system should normally be available during Resource Work Cooperative operating hours, subject to internet and hosting availability.
NFR-16	Operational Continuity	If the system is temporarily unavailable, staff shall use the existing paper forms and the records shall be entered into the system later by an authorized Manager/Administrator.
NFR-17	Testing	Core system functions shall be tested before the system is deployed for operational use.
NFR-18	Documentation	A simple user manual shall be provided for staff and administrators.
NFR-19	Training	User training and support shall be provided during the first week after implementation.

7.1. Usability

Usability is important because the system will be used during normal daily operations.

The interface should be simple and should avoid unnecessary steps. Staff should be able to enter information quickly without needing technical knowledge.

Where possible, the system should use:

- dropdown lists
- checkboxes or selection buttons
- number fields
- automatic calculations
- clear save and cancel buttons
- simple success and error messages

7.2. Device Support

The system is intended to be a tablet-friendly web application, which is part of the internship project scope.

The interface should work on:

- desktop computers
- tablets

The system does not need a separate mobile application for the first version.

7.3. Security

Security requirements should remain basic but appropriate for an internal operational system.

The first version should provide:

- authorized user access
- secure password handling if login accounts are used
- basic separation between staff and manager functions
- protection against unauthorized editing or deletion of records

Advanced security controls are outside the scope of the first version unless later required.

7.4. Reliability and Data Protection

The system should store submitted information correctly and prevent obvious invalid data from being saved.

Operational data should also be backed up using the selected database or hosting environment.

A complex disaster recovery system is not required for the initial version.

7.5. Maintainability

The solution should remain simple enough for future developers or IT staff to understand and maintain.

The project should avoid unnecessary technologies or complicated architecture.

Basic technical documentation should be provided for:

- system setup
- database structure
- deployment
- important configuration
- basic maintenance

7.6. System Downtime and Operational Continuity

If the system becomes temporarily unavailable due to an internet, server, or application issue, staff will continue using the existing paper datasheets to record operational activities.

Once the system becomes available again, the paper records will be entered into the system by an authorized Manager/Administrator.

This approach ensures that daily waste tracking activities can continue without depending completely on system availability.

Offline application functionality and automatic data synchronization are not required for the first version of the system.

7.7. Testing and User Training

The system will be tested before deployment to confirm that the main functions, calculations, validation rules, user access controls, and database operations work as expected.

Testing will include the main waste tracking workflows and will verify that records can be entered, calculated, stored, viewed, edited where permitted, and used for basic reporting.

After implementation, a simple user manual will be provided to Resource Work Cooperative.

Staff training and implementation support will also be provided during the first week after deployment to help users understand the main data-entry, record management, and reporting functions.

8. System Constraints and Assumptions

This section describes the main project limitations and the assumptions currently being used while the system is being designed.

8.1. Project Constraints

Table 20: Project Constraints

ID	Constraint
C-01	The system will be designed and developed by one intern.
C-02	The internship duration is 12 weeks, with limited remaining development time.
C-03	Development time is approximately 15-20 hours per week.
C-04	The project must focus on the most important waste tracking workflows rather than every possible operational process.
C-05	The system should remain simple enough to be completed, tested, deployed, and handed over within the internship period.
C-06	The solution should be suitable for a small organization and should avoid unnecessary complexity.
C-07	The system should support the current operational process rather than require major changes to how staff work.
C-08	Existing Google Workspace and reporting practices should be considered where practical.
C-09	Advanced integrations, automation, and enterprise-level features are outside the initial project scope unless identified as essential.

8.2. Current Assumptions

The following assumptions are being used for the first system draft. These should be confirmed with staff where required.

Table 21: Assumptions

ID	Assumption	Status
A-01	Detailed E-waste Pickup workflow, including final weight calculation and double-counting rules.	Pending clarification

A-02	Detailed Scrap fields and workflow.	Pending clarification
A-03	Detailed waste diversion and Monthly NRE calculation logic from the existing Excel workbook.	Pending analysis
A-04	Final dashboard charts and visualizations.	Pending analysis
A-05	Some other outbound waste processes are not included in the initial system unless later identified as necessary.	On hold
A-06	Existing Excel reports and graphs will be reproduced where practical rather than rebuilding every spreadsheet function.	Project assumption

8.3. Scope Control

Any new requirement identified during development should be reviewed before being added to the project.

If a new requirement would significantly increase development time or system complexity, it should either:

- replace a lower-priority feature,
- be simplified, or
- be recorded as a future enhancement.

This approach will help ensure that a complete and usable system can be delivered within the available internship period.

9. Acceptance Criteria

This section defines the main conditions that should be met for the first version of the Waste Tracking Operations System to be considered complete and usable.

The acceptance criteria focus on the main project goals rather than every possible feature.

Table 22: Acceptance Criteria

ID	Acceptance Criteria
AC-01	Staff can access the web application using a desktop or tablet browser.
AC-02	Staff can enter data for the main waste tracking workflows included in the project scope.
AC-03	Entered records are stored successfully in a central database.
AC-04	The system validates required fields and basic data rules before saving records.

AC-05	Material percentage entries are checked to ensure they total 100% where required.
AC-06	The system correctly deducts configured container tare weights where required.
AC-07	The system correctly calculates material weights from entered percentages where required.
AC-08	The system calculates estimated whitegoods weight using 48 kg per recorded whitegoods.
AC-09	Staff can review and correct data before final submission.
AC-10	After submission, only Manager/Admin users can edit or delete records.
AC-11	Users can view previously entered records.
AC-12	Manager/Admin users can view basic reports and summaries based on stored data.
AC-13	Report data can be exported in Excel or CSV format where required.
AC-14	The system reduces the need to manually transfer the same operational data from paper forms into Excel.
AC-15	The main data-entry forms are simple enough for staff to use with minimal training.
AC-16	The system is deployed in an environment where Resource Work Cooperative staff can access and test it.
AC-17	Basic user and technical documentation is provided for system use and handover.
AC-18	Core system functions are tested successfully before deployment and final handover.
AC-19	Manager/Admin users can view a daily summary of the main operational activities recorded in the system.
AC-20	A user manual is provided to Resource Work Cooperative.
AC-21	Initial user training and support are provided after system deployment.
AC-22	A documented paper-based fallback process is available for temporary system downtime.

9.1. Minimum Successful Delivery

The project will be considered successfully delivered if a working version of the system can:

- capture the main waste tracking data digitally,
- store the data in a structured database,
- allow records to be viewed and corrected,
- provide basic reporting or summaries,
- work on common devices,
- and be deployed for staff use.

Not every current spreadsheet calculation or operational process needs to be fully automated in the first version.

Any lower-priority requirements that cannot be completed within the internship period may be moved to future development.

10. Future Enhancements

The first version of the Waste Tracking Operations System will focus on the main requirements identified for the internship project.

The following features may be considered for future development:

- More advanced waste diversion and sustainability dashboards
- Additional reporting and automated regulatory reports
- Direct integration with weighing equipment
- Upload and storage of weighbridge dockets and supporting documents
- More detailed electrical item and repair tracking
- Additional outbound waste and recycling workflows
- Advanced user roles and permissions
- Improved mobile support and possible offline data entry
- Integration with other systems used by Resource Work Cooperative

Future enhancements should be considered after the first version of the system has been deployed and evaluated by staff.